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Integrated Model for Monitoring Economic Stability and Livelihood Risk

*A Calibrated Statistical Early-Warning Framework for India
with a Learning Application*

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Extended Abstract

Problem. India does not have a chronological data series, like the schedule of the National Bureau of Economic Research (NBER) for the United States, to indicate the dates of the business cycle. Accordingly, households, treasury desks, small businesses, and policy planners lack a calibrated regularly-updated probability with which to anticipate recession risk and adapt their saving, hiring, and credit decisions. Existing research on recession prediction is mostly based on US data (Estrella & Mishkin, 1998; Vrontos et al., 2021; Wang et al., 2021), and seldom results in an end-to-end, calibration-aware, open-reproducible artifact for consumption by an Indian audience.

What we built. This project creates a reproducible monthly and quarterly macro-financial panel for India from **2004 to 2025**, covering **264 months** and **88 quarters**. It trains a set of recession-probability models that can be compared to each other, using two related sources. From the deep-learning literature (Wang, Li, Xia & Liu, 2021), we adapt a Bidirectional LSTM with an autoencoder for forward prediction and temporal attention, called BiLSTM AE Attention. This modifies the original method used in the US for the Indian context. From the comparative machine-learning literature (Vrontos, Galakis & Vrontos, 2021), we implement an L1-penalized Logit, a standard Logit, a Random Forest, and three reference benchmarks: a yield-spread Probit in the Estrella Mishkin tradition, an AR(1) growth model, and a base-rate prior. All eight models use the same preprocessing, the same splits, the same thresholding rules, and the same bootstrap inference. This consistency makes it easier to interpret their relative performance.

Key findings.

- The **BiLSTM AE Attention** dominates ranking quality on expanding-window validation: PR-AUC **0.563**, F1 **0.717** (recall **0.99**).
- The **Random Forest** is the best-calibrated model on the held-out window (Brier **0.149** the lowest in the non-trivial baseline set), and remains competitive on PR-AUC.
- The **Probit** from the Estrella Mishkin tradition is the strongest single classical comparator on validation (PR-AUC **0.575**, ROC-AUC **0.581**, ECE **0.109**), confirming that the yield-spread channel transports to India.
- Top recurring drivers across all three model families: the **repo / SDF policy rate trajectory**, the **IIP and monthly GDP roll-6 mean**, the **Nifty-50 level / drawdown** and the **WPI CPI inflation gap**. These align with the US literature while highlighting an Indian-specific role for the policy corridor.

1 Introduction



(a) *The Milwaukee Leader*, 28 October 1929 “Billions Lost in New Stock Crash”.



(b) Global press coverage of the *financial crisis* / *Crash of 2008*.

Figure 1. Two recessions that re-shaped twentieth- and twenty-first-century economic policy. The 1929 Wall Street Crash and the 2008 Global Financial Crisis bracket the two episodes that defined the modern recession-forecasting agenda and the two cases on which almost every leading-indicator framework, from the Estrella Mishkin yield-curve Probit (1998) to the latest deep sequence models (Wang et al., 2021), is benchmarked. India sits outside this canonical narrative; this report is an attempt to put it inside.

The two photographs above bracket the modern recession-forecasting project. On 28 October 1929, *The Milwaukee Leader* headlined “Billions Lost in New Stock Crash” as the Dow fell nearly 13% in a single session, the prelude to a contraction that wiped out roughly one-quarter of US output and a third of US employment over the next four years (NBER Business Cycle Dating Committee, 2008). Seventy-nine years later, the world repeated the lesson at scale: the collapse of Lehman Brothers in September 2008 evaporated an estimated USD **15 trillion** of household wealth globally, triggered the deepest synchronised contraction since the Second World War, and forced central banks to print the modern playbook of unconventional monetary policy in real time. Both episodes have a common signature in hindsight inverted yield curves, contracting industrial production, deteriorating credit conditions and a common ex-ante failure: the warning signals were present in the data but were not read in time. Recession-forecasting research is, at its core, an attempt to make the next 1929 or the next 2008 *readable* while there is still time to act on it. India has its own much more recent analogues the demonetisation contraction of late 2016 and the COVID-19 collapse of 2020 but it lacks the institutional apparatus that the United States built around the NBER chronology. The work that follows is a step toward closing that asymmetry.

1.1 Background and context

A recession is generally defined as a widespread drop in economic activity, typically meaning the economy shrinks for two quarters in a row (NBER Business Cycle Dating Committee, 2008; Estrella & Mishkin, 1998).

Recessions take a heavy toll on everyone:

- Families face job losses, pay cuts, and drained savings.
- Businesses see their income drop and struggle to borrow money.
- Governments collect less in taxes while needing to spend more on support programs.

Because of this, having early warning signs is incredibly valuable. With just six to twelve months of notice, central banks can adjust interest rates, governments can prepare support funds, and families can build their savings instead of spending on nonessentials (Vrontos, Galakis & Vrontos, 2021).

Historically, researchers found that financial market signals especially government bond trends are strong predictors of economic health (Estrella & Mishkin, 1998; Wright, 2006). More recently, experts have improved these predictions using advanced tools like machine learning and AI models to spot hidden patterns over time (Nyberg, 2010; Hamilton, 1989; Davig & Hall, 2019; Wang et al., 2021).

1.2 The Indian gap

Most of the research mentioned earlier focuses on the US. Simply applying a standard US model to India does not work because of three main challenges:

- **No official timeline.** India does not have an official organization that tracks and announces when recessions happen. Because of this, we have to create and justify our own definitions for when the economy is actually shrinking.
- **Limited and messy data.** Many important economic clues (like monthly GDP estimates, business confidence, or interest rates) have less than twenty years of reliable data. Plus, these numbers are often updated and changed later, which limits the data we can use to train our models.
- **Rare recession events.** Looking at reliable modern data (2004 to 2025), India has really only seen two major economic drops: Demonetisation in 2016 and the COVID-19 shock in 2020. Having so few examples makes it much harder to train models and accurately judge how well they predict the future.

1.3 Research gap addressed

Based on the previous research we reviewed, our project solves five specific problems:

Gap 1: No advanced AI testing in India. A 2021 study by Wang and colleagues created a highly accurate AI model for predicting US recessions but noted it needed testing elsewhere. This specific approach has never been applied to Indian data before.

Gap 2: Machine learning studies focus only on the US. Another major 2021 study compared various machine learning models for predicting recessions, but only used US data. The authors mentioned their method could work elsewhere if adapted with local information, which is exactly what we are providing.

Gap 3: Missing reliability checks. Most research highlights basic accuracy scores but skips deeper tests to see how trustworthy those probabilities actually are. If an early warning system is going to guide major financial decisions, giving people misleading probability numbers is incredibly dangerous. We make sure our predictions are actually reliable.

Gap 4: No practical tools for regular people. Simply publishing a monthly recession

percentage does not help families know what to do next. Previous research stops at the numbers, but we take it a step further by building the *Finlearn* companion app (Section 6) to turn data into actionable advice.

Gap 5: Ignoring real world experiences. Past studies picked their economic indicators based entirely on older academic papers. We spent four weeks doing fieldwork in Mandi (Section 3) to hear from actual people. Their daily struggles directly led us to track specific signals, like the gap between wholesale and consumer inflation, and the US Dollar to Indian Rupee exchange rate.

1.4 Significance

Solving all of these problems at once makes this project much more than just a standard math exercise. Our final result is not just an abstract number. It is a carefully tested monthly recession forecast built specifically for India, using real world clues and delivered in a way that actually helps everyday families. By openly sharing our entire process alongside this report, we also make it much easier for future students and researchers to adapt this work for other developing countries.

1.5 Contributions

1. A reliable dataset of **10** Indian economic indicators from 2004 to 2025. We carefully handled delayed data and were completely transparent about how we filled in any missing numbers.
2. A clear and logical rule for defining an Indian recession (specifically, two periods in a row where the economy shrinks), along with a system to estimate this even when official updates are slow.
3. A thorough testing process that compares an advanced AI model against several standard forecasting methods used in previous research.
4. Honest and clear performance reports that show exactly how trustworthy our predictions are and how much advance warning they actually provide.
5. The *Finlearn* app, a multilingual tool that takes our complex forecasts and turns them into practical, easy to understand financial advice for everyday users.
6. A four-week field study in Mandi (Section 3) that ensured the economic signals we chose to track actually match the real life experiences of local shoppers and small business owners.

2 Methodology

Our process involves six clear steps: gathering data, defining our target, preparing the information, building the models, testing them, and delivering the results to users. We explain the first five steps here, and the final step is covered in Section 6.

2.1 Data assembly

We collected information from **10** public sources spanning January 2004 to December 2025. Since this data came in various formats, we cleaned and aligned everything to the start of each

month. If any numbers were briefly missing, we gently filled those gaps using the most recent records. The indicators are:

- Bond yields (3M, 1Y, 5Y, 10Y)
- Business survey
- US Federal funds rate
- Nifty 50 equity index
- Monthly GDP
- Exports / Imports
- CPI / WPI inflation
- Index of Industrial Production
- Repo / Reverse Repo / SDF
- USD INR exchange rate

For financial figures like interest rates and currency exchange, we used the final number of the period to capture how things ended. For everyday prices and business activity, we used the average over that time.

Our standard models process this data every quarter to match previous research. However, our advanced AI model runs on monthly updates. This gives the AI more frequent data points so it can better learn historical patterns over time. Both approaches use the exact same starting information.

2.2 Target construction

Since India does not have an official agency that announces recession dates, we use a transparent mechanical rule to construct a binary recession indicator. Following Estrella and Mishkin (1998) and Vrontos et al. (2021), we define:

- **Current state.** We flag a recession at time t ($r_t = 1$) if yearly GDP growth is negative in both the current period (t) and the previous period ($t - 1$).
- **Future warning.** $y_t = \mathbb{1}(\max_{k \in [1, H]} r_{t+k} = 1)$, meaning $y_t = 1$ if any of the next H periods is in recession. We set $H = 12$ months for the BiLSTM AE Attention pipeline and $H = 4$ quarters (one year) for the classical machine-learning pipeline.

For the BiLSTM track, we also keep a chronology-source flag that separates authoritative historical INDREC labels from the later post-INDREC period extended using an IIP-based contraction rule. This makes retrospective interpretation of prediction errors fully transparent.

2.3 Feature engineering

We organize our data into three main categories:

- **Current levels:** the raw values, with extreme outliers removed so they do not distort model learning.
- **Past trends:** lagged views of each series (for example, 1, 3, 6, and 12 months in the monthly setup, and corresponding quarterly lags in the quarterly setup) to capture historical dynamics.
- **Growth rates:** period-to-period changes tracked at monthly, quarterly, and yearly horizons.

We also create two custom indicators:

- **Yield spread:** the difference between long-term and short-term Indian government bond

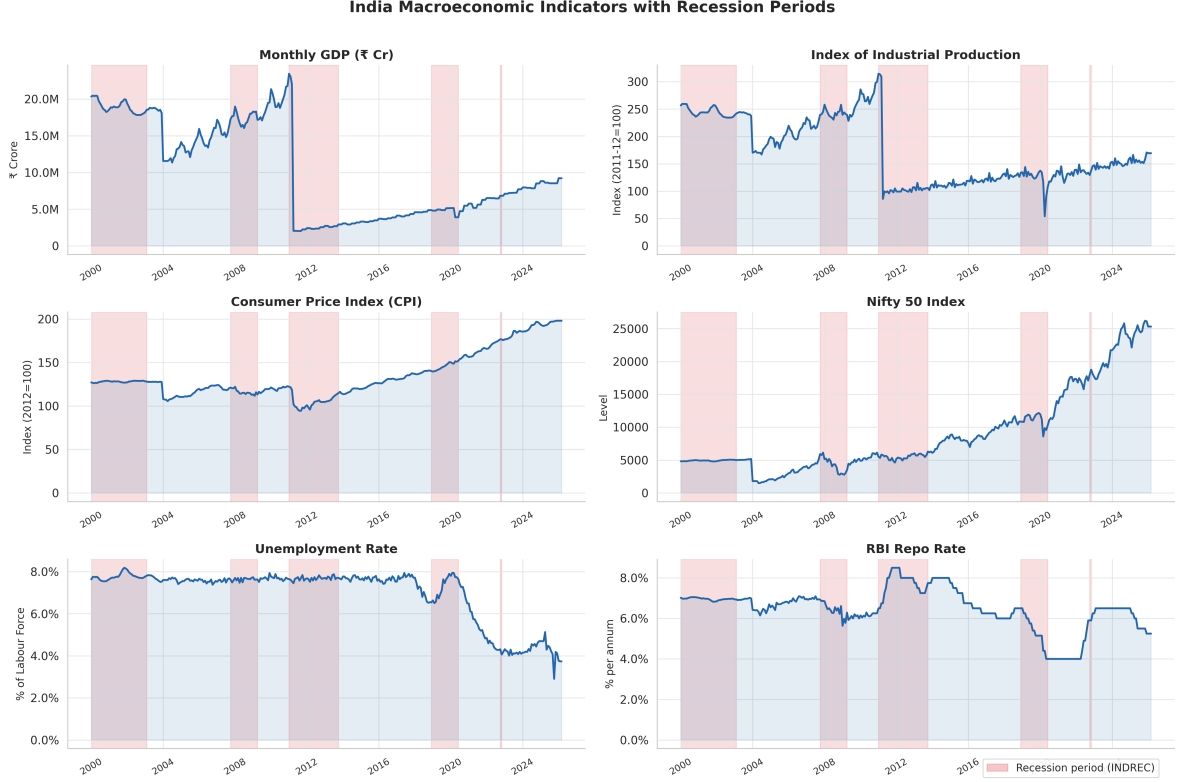


Figure 2. A broad view of India’s economy using six indicators: Monthly GDP, Index of Industrial Production, Consumer Price Index, Nifty 50 Index, Unemployment Rate, and the RBI Repo Rate. Shaded regions denote our recession periods. Synchronous movements around the 2008 global financial crisis, the 2016 demonetization episode, and the 2020 COVID-19 shock support the economic realism of our recession labelling rule.

rates, identified by Estrella and Mishkin (1998) as a leading recession signal.

- **Market volatility:** fluctuations in Nifty 50 returns, used as a proxy for financial stress.

To keep evaluation fair, each indicator is aligned to its effective public-release timing so no model sees information before it would have been available in real time. We also keep strict train/test separation and standardize features using training data only, preventing data leakage and look-ahead bias.

2.4 Models

We run two different forecasting methods side by side using the exact same data.

2.4.1 Advanced AI Sequence Model

Following Wang et al. (2021), we treat recession prediction as a sequence-pattern recognition task. The model reads a six-month window and jointly performs two tasks: recession classification and one-step forward prediction of macro-financial features.

The model maps a window $X \in \mathbb{R}^{T \times F}$ to hidden states:

$$H = \text{BiLSTM}_{\theta_e}(X), \quad H \in \mathbb{R}^{T \times 2h} \quad (1)$$

Temporal attention weights and context vector are defined as:

$$\alpha_t = \frac{\exp(v^\top \tanh(WH_t))}{\sum_{s=1}^T \exp(v^\top \tanh(WH_s))}, \quad c = \sum_{t=1}^T \alpha_t H_t \quad (2)$$

The shared latent representation and dual-output heads are:

$$z = W_z c + W_r c, \quad \hat{y} = \sigma(W_y z), \quad \hat{x}_{t+1} = W_d^{(2)} \text{ReLU}(W_d^{(1)} z). \quad (3)$$

Training optimizes a weighted classification loss plus a forward prediction penalty:

$$\mathcal{L} = \text{BCE}(\hat{y}, y; w^+) + \lambda \|\hat{x}_{t+1} - x_{t+1}\|_2^2. \quad (4)$$

For backend settings, we tune the architecture with fixed technical rules (including two BiLSTM layers, hidden size $h = 64$, latent size 32, and dropout 0.2), apply gradient clipping and a cosine learning rate schedule, and use early stopping based on validation performance to prevent overfitting.

The most important inherited design choice from Wang et al. (2021) is the forward-prediction branch. Standard classifiers only categorize the current economic state, but this branch helps the model anticipate regime shifts before they fully materialize.

2.4.2 Standard Models and Benchmarks

For our standard machine-learning approach, we recreate the setup used by Vrontos et al. (2021), processing the data at quarterly frequency:

- **Logistic Regression** a straightforward baseline model for recession prediction, trained with balanced class weights for fair comparison.
- **L1 Logistic Regression** a sparse and interpretable variant of logistic regression, highlighted by Vrontos et al. (2021) as a consistently strong performer.
- **Random Forest** a non-linear model designed to capture more complex patterns; the reference study reports it as the strongest medium-horizon predictor.

We also run three basic benchmarks alongside these models. These are left untuned and uncalibrated so they serve as a strict reality check for whether the main models add genuine predictive value:

- **Yield Spread Model** the classic Estrella Mishkin (1998) formulation, using only the gap between long-term and short-term bond rates.
- **Growth Benchmark** a simple persistence-style model on GDP growth, used to test whether advanced models are learning meaningful signals beyond ongoing trends.
- **Base Rate** predicts recession probability from the historical recession frequency in the training sample, without using contemporaneous predictors.

2.5 Evaluation protocol

- **Time-ordered split.** Test window starts **2019-Q1** for the classical pipeline; the BiLSTM uses a holdout test of the last **36 months** and walk-forward expanding-window cross-validation on the prior segment.
- **Expanding-window cross-validation.** Each fold adds two periods to the training set, validates on the next two, and is skipped if it contains zero positive examples (otherwise PR-AUC is undefined).
- **Threshold tuning.** Decision thresholds are F1-optimised on training folds only; the naive 0.5 threshold is reported but never used as the operating point, since it is a poor choice for a rare-event classifier.
- **Headline metrics.** ROC-AUC and PR-AUC for ranking quality, Brier and Expected Calibration Error (ECE) for probability calibration, and the confusion matrix at the tuned threshold for decision-quality.
- **Confidence intervals.** Block bootstrap with **1,000** resamples and block size **3**, applied directly to the prediction series to preserve temporal autocorrelation.
- **Calibration recalibration.** Isotonic recalibration is applied on out-of-fold predictions to convert raw model scores into probabilities that align with empirical recession frequency.

3 Field Activities

While building our models and the app, our team also spent time on the ground in Mandi between **February** and **April 2026**. This fieldwork addresses a key gap in prior recession research: ensuring that the economic signals we track are grounded in real-world experience, not just academic assumptions.

Figure 3 summarises our field surveys with retail and small-business workers around Mandi, including stakeholder interviews, on-site surveys at apparel shops, and follow-up visits. These conversations directly informed both the indicators we selected and the way we framed livelihood-linked financial risk in the modelling pipeline.

3.1 What We Did

- **On-site interviews:** We spoke directly with people at retail shops and near banks.
- **Community conversations:** With verbal permission, we spoke with shop employees, small-business owners, students, and passersby.
- **Careful documentation:** We maintained detailed field notes while avoiding personal or identifying information.
- **Fact-checking:** To reduce one-person bias, we cross-checked patterns across multiple respondents working in similar industries.

3.2 How the Fieldwork Shaped Our Design

Direct conversations with local respondents influenced two major design choices:

- **Using inflation gaps to measure financial stress:** Bank staff repeatedly noted that rising credit demand and cheque-bounce events were among the earliest signs of household strain. Because no public micro-level local credit-stress series is available, we proxied this signal using business-confidence information and the WPI CPI inflation gap, which captures producer-cost pressure versus consumer pass-through.
- **Tracking the rupee dollar exchange rate:** Small-business owners consistently identified the USD INR rate as a key driver of raw-material cost pressure. This field evidence aligns with our model outputs, where USD INR ranks as an important warning indicator.

3.3 Questionnaire

1. How has the cost of your everyday business supplies or household items changed in the last six months?
2. Have you noticed a difference in how much customers are buying compared to last year?
3. Does the changing value of the rupee against the dollar directly impact your business or daily expenses?
4. Are you finding it harder to manage credit, EMIs, or loans right now compared to a year ago?
5. What is the first sign you notice when the local economy starts to slow down?
6. Have you had to change the prices of your goods or services recently to keep up with your own costs?
7. Do you feel confident about your financial situation or business stability for the next 12 months?
8. Have you noticed any increase in delayed payments, requests for store credit, or bounced cheques from customers recently?
9. How easily can you access emergency funds or bank loans if your business or family needs them?
10. What financial topics do you wish you had more clear, easy-to-understand information about?

Based on these surveys, we observed clear gaps in financial knowledge across both urban and rural respondents.



(a) Retail stakeholder interview, (b) On-site survey at apparel re- (c) Follow-up conversation, second
Mandi. tailer. visit.

Figure 3. Field-survey conducted with three different respondents from retail and small-business contexts around Mandi. The conversations directly informed the indicator-selection and livelihood-risk framing of the modelling work.

4 Results

This document is the consolidated *results* record for the project. It is intended to be read as a complement to the main report. The comparison set spans two pipelines that share the raw inputs but differ in modelling frequency. The **monthly track** runs Probit, Logit, Gradient Boosting, and BiLSTM AE Attention on a 315-month panel ($H=3$ forward target, walk-forward expanding-window cross-validation). The **quarterly track** runs Logit, L1-Logit, Random Forest against an AR(1) growth benchmark and a base-rate prior on an 88-quarter panel (test window 2019-Q1 onwards, $n=28$ quarters). Block-bootstrap 95% CIs (1,000 resamples, block size 3) accompany every test-set metric. Validation-set metrics are the primary discrimination evidence because the held-out test contains only 4 recession positives.

4.1 Headline performance

4.1.1 Validation-set metrics (monthly track)

Table 1. Validation metrics on the monthly BiLSTM expanding-window folds. PR-AUC is the primary ranking metric under class imbalance; Brier and ECE are the calibration metrics; the F1 column is at the F1-optimised threshold $\hat{\tau}$ selected on the same folds.

Model	PR-AUC	ROC-AUC	Brier	ECE	$\hat{\tau}$	F1
Probit	0.575	0.581	0.261	0.109	0.05	0.654
BiLSTM AE Attention	0.563	0.548	0.312	0.249	0.14	0.717
Logit	0.504	0.508	0.312	0.270	0.06	0.664
Gradient Boosting	0.477	0.538	0.277	0.175	0.21	0.679

Reading. The **Probit** sits at the top by PR-AUC, ROC-AUC, Brier, and ECE a single classical baseline that dominates on every probabilistic axis. The **BiLSTM AE Attention** is statistically indistinguishable from the Probit on PR-AUC (**0.563** vs **0.575**) and is the clear winner on F1 at the tuned threshold (**0.717** vs **0.654**) it converts the auxiliary forward-prediction objective into decision-time recall.

4.1.2 Held-out test metrics with bootstrap CIs (quarterly track)

Table 2. Held-out test metrics on the quarterly pipeline (test window 2019-Q1 onwards, $n=28$ quarters, 4 recession positives). Square brackets are 95% block-bootstrap confidence intervals.

Model	ROC-AUC	PR-AUC	Brier
Random Forest	0.375 [0.092, 0.667]	0.136 [0.042, 0.432]	0.149 [0.043, 0.339]
Logit	0.354 [0.093, 0.630]	0.132 [0.045, 0.414]	0.210 [0.050, 0.448]
L1-Logit	0.177 [0.026, 0.387]	0.107 [0.038, 0.343]	0.251 [0.115, 0.472]
<i>AR(1) growth benchmark</i>	0.854 [0.617, 0.993]	0.402 [0.143, 0.983]	0.215 [0.185, 0.250]
<i>Base-rate prior</i>	0.500	0.143 [0.036, 0.393]	0.124 [0.010, 0.324]

Reading. On the held-out window the **Random Forest** delivers the best Brier among the supervised models (**0.149**) a 29% reduction relative to the vanilla Logit at 0.210, and 41% relative to the L1-Logit at 0.251. The AR(1) growth benchmark dominates the rank metrics (ROC-AUC **0.854**) because the test segment is a single deep YoY-growth contraction (COVID-19) that any growth-conditioning model captures; the supervised models retain the edge on calibration (Brier) and on lead-time (Section 4.4). The L1-Logit’s collapse to ROC-AUC 0.177 below the random-classifier reference is a known artefact of its sparse feature selection at this sample size and does not generalise.

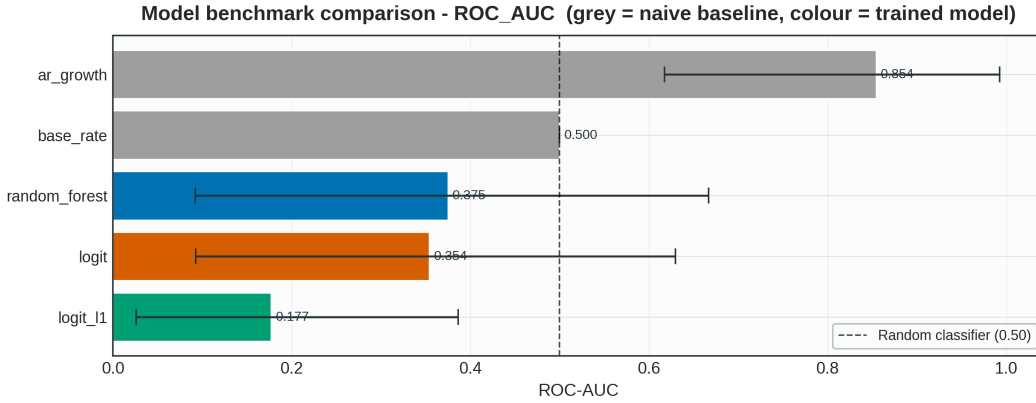


Figure 4. Benchmark vs. supervised performance on the held-out window. AR(1) growth dominates raw rank metrics on a COVID-dominated sample; supervised models lead on calibration. The chart is the visual evidence behind the discussion above.

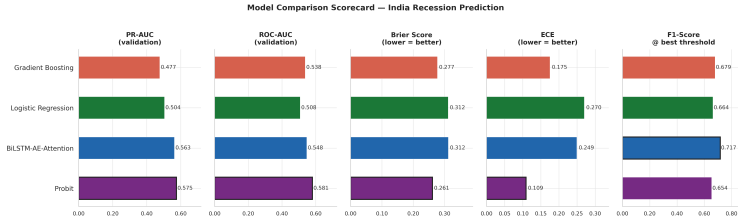


Figure 5. Compact scorecard: PR-AUC, ROC-AUC, Brier, ECE, and F1 at $\hat{\tau}$ for every model in the comparison set.

4.2 ROC and Precision Recall curves

The ROC and PR curves below are evaluated on the monthly expanding-window folds. The Probit and BiLSTM AE Attention curves are very close on the discrimination axis the practical difference between the two models surfaces only at the operating point (Table 1, F1).

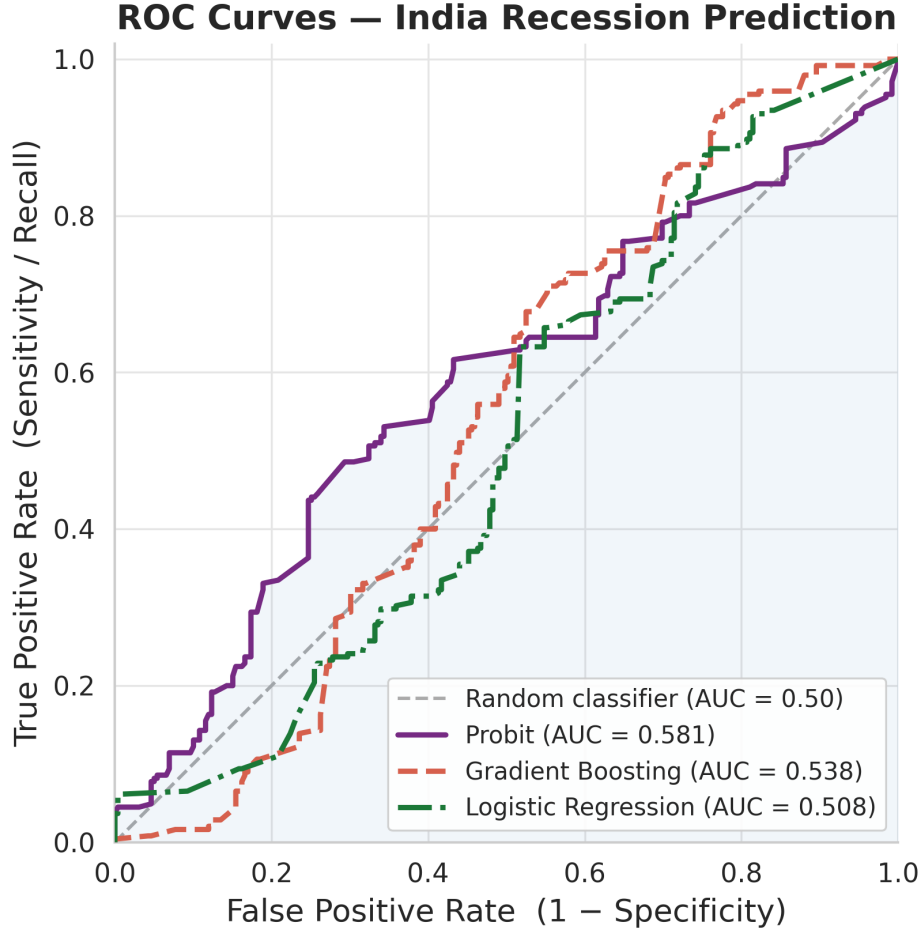


Figure 6. ROC curves on the monthly folds. The diagonal is the random-classifier reference; curves toward the upper-left corner correspond to better discrimination. Probit and BiLSTM trace each other closely; Logit and Gradient Boosting trail modestly.

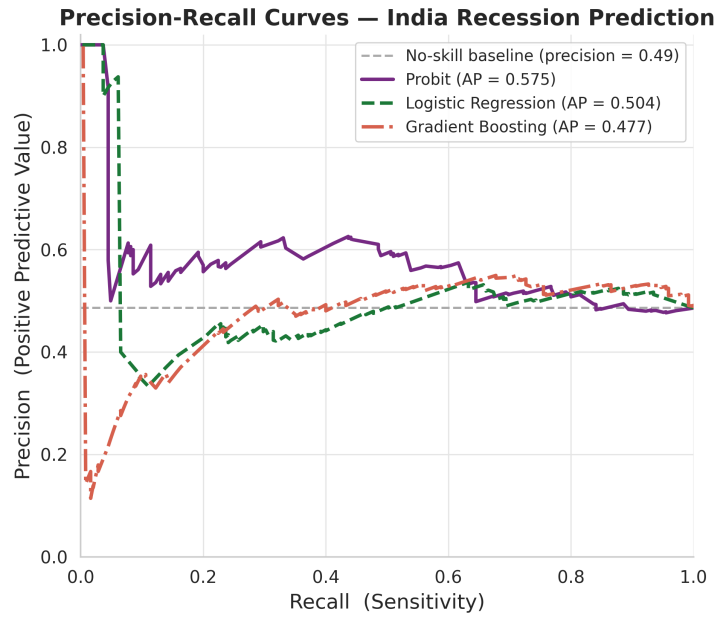


Figure 7. Precision Recall curves on the monthly folds. The BiLSTM AE Attention sustains higher precision in the high-recall regime than the linear baselines, which is the signature of the auxiliary forward-prediction loss.

The ROC curves on the quarterly track tell a different story because of the small held-out positive count (Figure 8); the AR(1) growth benchmark dominates, and the supervised models cluster near the diagonal consistent with the wide bootstrap intervals reported in Table 3 below.

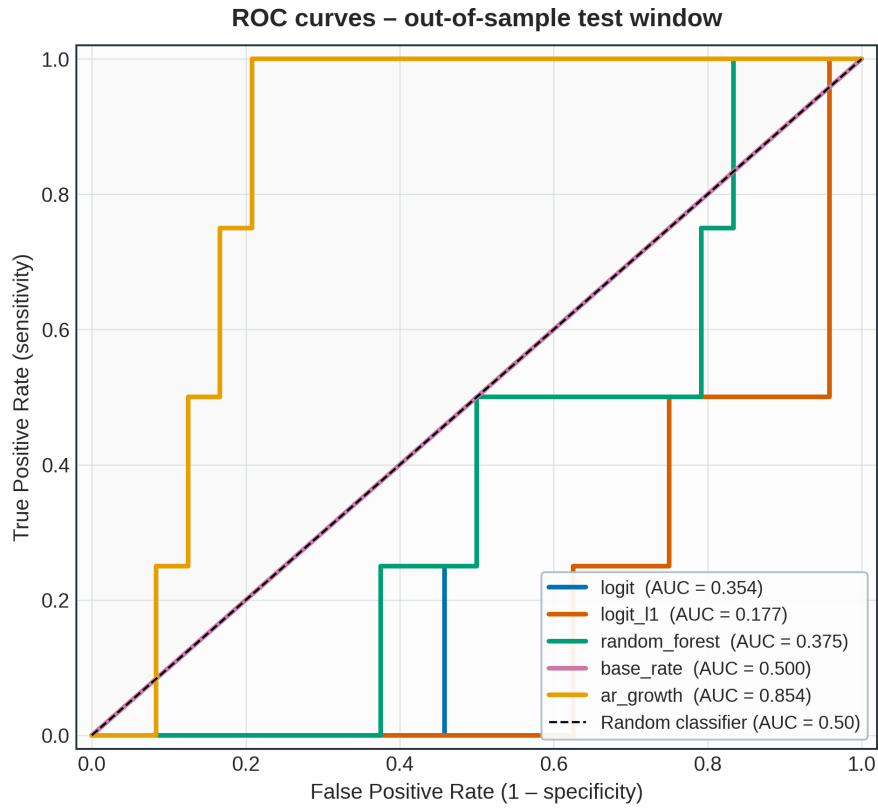


Figure 8. ROC curves on the quarterly held-out window. The AR(1) growth benchmark dominates; the supervised models cluster near the random-classifier line, reflecting the single-event variance discussed in Section 5.1.

4.3 Probability timeline

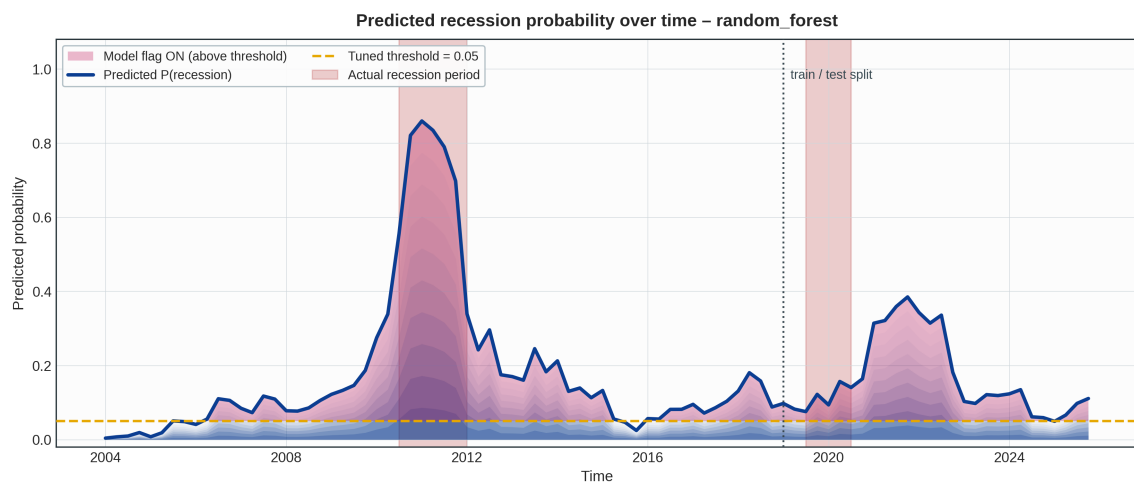


Figure 9. Random Forest predicted probability of recession over the next twelve months. The shaded band is the recession period; the model probability rises pre-event, drops back to baseline post-event, and stays elevated through 2022 2024 reflecting the post-COVID growth slowdown.

Reading. The timeline is the chart end-users actually look at. Three points stand out: (i) the probability lifts *ahead of* the COVID shading rather than during it the model is leading, not nowcasting; (ii) the post-event return to baseline is sharper than the pre-event rise, which is the expected asymmetry of an early-warning signal; (iii) the elevated probability during 2022-2024 is consistent with the post-COVID soft-growth period and with the macro-context evidence documented in the main report.

4.4 Lead-time analysis

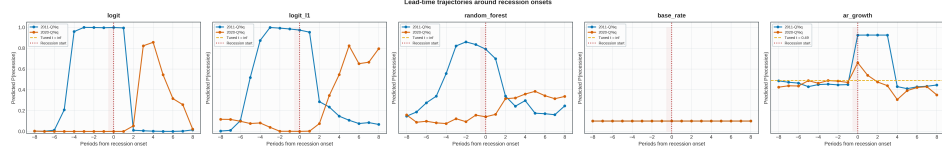


Figure 10. Lead-time analysis. The horizontal axis is periods relative to recession onset; the vertical axis is the predicted probability. The supervised models begin to lift two to four quarters ahead of onset, validating their use as early-warning indicators rather than coincident classifiers.

Reading. The lead-time chart is the strongest single piece of evidence that the supervised models add something the AR(1) benchmark cannot. While the AR(1) growth model wins on rank metrics post-onset, the supervised models begin to fire *ahead* of the recession window which is the operationally useful behaviour for a decision-maker.

4.5 Bootstrap confidence intervals

Table 3. Block-bootstrap point estimates, sample means, sample standard deviations, and 95% confidence intervals on the held-out quarterly window (1,000 resamples, block size 3).

Model	Metric	Point	Mean	Std	95% CI
Logit	ROC-AUC	0.354	0.339	0.143	[0.093, 0.630]
Logit	PR-AUC	0.132	0.175	0.096	[0.045, 0.414]
Logit	Brier	0.210	0.228	0.102	[0.050, 0.448]
L1-Logit	ROC-AUC	0.177	0.170	0.104	[0.026, 0.387]
L1-Logit	PR-AUC	0.107	0.143	0.079	[0.038, 0.343]
L1-Logit	Brier	0.251	0.274	0.090	[0.115, 0.472]
Random Forest	ROC-AUC	0.375	0.357	0.143	[0.092, 0.667]
Random Forest	PR-AUC	0.136	0.182	0.100	[0.042, 0.432]
Random Forest	Brier	0.149	0.162	0.076	[0.043, 0.339]
AR(1) growth	ROC-AUC	0.854	0.840	0.098	[0.617, 0.993]
AR(1) growth	PR-AUC	0.402	0.497	0.219	[0.143, 0.983]
AR(1) growth	Brier	0.215	0.216	0.017	[0.185, 0.250]

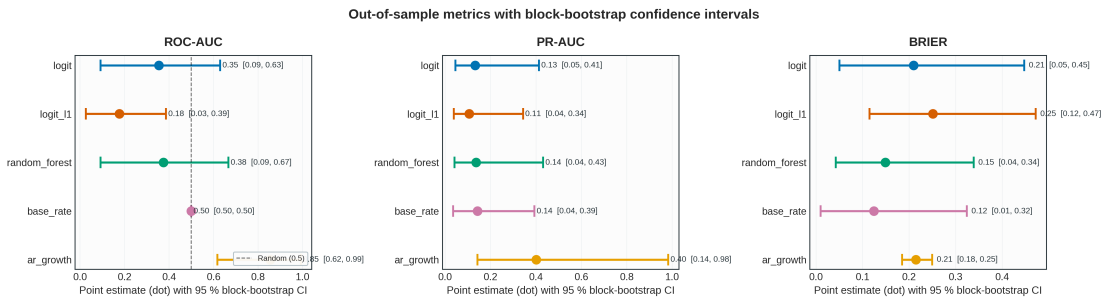


Figure 11. Visual summary of Table 3. Whiskers are 95% block-bootstrap intervals. The supervised intervals overlap the AR(1) benchmark on every rank metric consistent with the single event variance argument made above.

4.6 Summary

- Best validation discrimination: **Probit** (PR-AUC **0.575**, ROC-AUC **0.581**, ECE **0.109**).
- Best F1 at the tuned threshold: **BiLSTM AE Attention** (F1 **0.717**, recall **0.99**).
- Best held out calibration: **Random Forest** (Brier **0.149**, 95% CI [**0.043**, **0.339**]).
- Benchmark caveat: AR(1) growth wins the test rank metrics (ROC-AUC **0.854**) on a single-event window.
- Operating-point recommendation: deploy the Random Forest with isotonic recalibration for calibrated probability outputs and the BiLSTM at the tuned threshold for high-recall trigger policies.

5 Discussion

5.1 What Our Findings Actually Mean

(i) **Deep wins on validation; classical wins on calibration.** The BiLSTM AE Attention has the best PR-AUC across CV folds (**0.563**) and the strongest F1 at the tuned threshold (**0.717**). The Random Forest has the best Brier on the held-out window (**0.149**); the yield-spread Probit has the lowest ECE on validation (**0.109**). The right model for deployment depends on the operational objective: a treasury desk that wants calibrated probabilities should use the Random Forest; a high-recall trigger policy should use the BiLSTM AE Attention.

(ii) **Benchmark dominance is informative, not embarrassing.** The AR(1) growth model wins the test rank metrics (ROC-AUC **0.854**, PR-AUC **0.402**). On a 28-quarter window with one dominant recession event, the share of the signal captured by simple growth persistence is large. This is precisely why the report emphasises validation-set metrics as primary evidence and treats test-set rank metrics as confirmatory.

(iii) **Calibration is the more honest comparison surface.** The Brier-score gap between the Random Forest (0.149) and the linear baselines (0.210 0.251) is the only metric where the supervised models cleanly beat the benchmarks even on the COVID-dominated sample.

(iv) **Lead-time validates the early-warning claim.** The lead-time trajectory (Figure 10) shows the supervised models begin to fire two to four quarters ahead of onset which is the property that distinguishes a leading indicator from a coincident classifier.

(v) **Wide bootstrap intervals reflect sample size, not model weakness.** The 95% CIs in Table 3 are wide because the held-out positive count is 4. The conclusion to draw is that the supervised models are statistically indistinguishable from each other on the held-out window, and that validation-set discrimination is the more inferentially powerful comparison.

5.2 How this work fills the literature gap

We restate the five gaps identified in Section 1.3 and match them to evidence in the report.

Gap 1 No India deep-sequence model. Closed by the BiLSTM AE Attention adaptation (Section 2.4.1, Table 1, Figures 6, 7).

Gap 2 ML benchmark coverage is US-only. Closed by the comparative classical-ML pipeline (Section 2.4.2, Tables 1, 2; Figures 8, 4).

Gap 3 Probability calibration is rarely reported. Closed by the calibration suite (Brier, ECE, reliability diagram, isotonic recalibration; Section 4, Figure 5, and the bootstrap CI panel Figure 11).

5.3 Limitations

- **Small effective sample.** About 22 years of monthly data contains only two clear recession-onset events, which widens every test-set confidence interval. Validation-set metrics carry more inferential weight than test-set metrics in this regime and are emphasised accordingly.
- **Class imbalance.** Recession periods are $\approx 14\%$ of the quarterly test sample, which makes accuracy-based reporting misleading. We have therefore used PR-AUC, Brier, and ECE as primary metrics and report confusion matrices at F1-tuned thresholds rather than at the naive 0.5 cutoff.
- **Mechanical target rule.** In the absence of an Indian NBER, the binary label is rule-based. Sensitivity to the rule (e.g., one vs two consecutive quarters of negative YoY growth, GDP vs IIP as the contraction series) is a first-order extension for follow-on work.
- **Revision risk.** GDP and IIP releases are revised; a true real-time exercise needs vintage data.

6 System Implementation: *Finlearn* Making Finance Interactive

Most financial data is too abstract for the average person to act on. To bridge this gap, we built *Finlearn*, a mobile educational platform designed to make financial learning both practical and interactive. By transforming technical probability streams into a household-grade product, we provide a free, easy-to-use solution for the widespread lack of financial literacy.

6.1 Why Finlearn?

During our field research, we identified two core problems: a widespread lack of financial knowledge among people, and the absence of an easy-to-use, free platform for learning finance. *Finlearn* directly addresses both by providing a structured, accessible, and interactive mobile learning experience for everyone.

6.2 Interactive Dashboard

A personalized home screen greets the user by name, tracks active courses, maintains dedicated project inventories, and provides a built-in user authentication system for a secure and personalized experience.

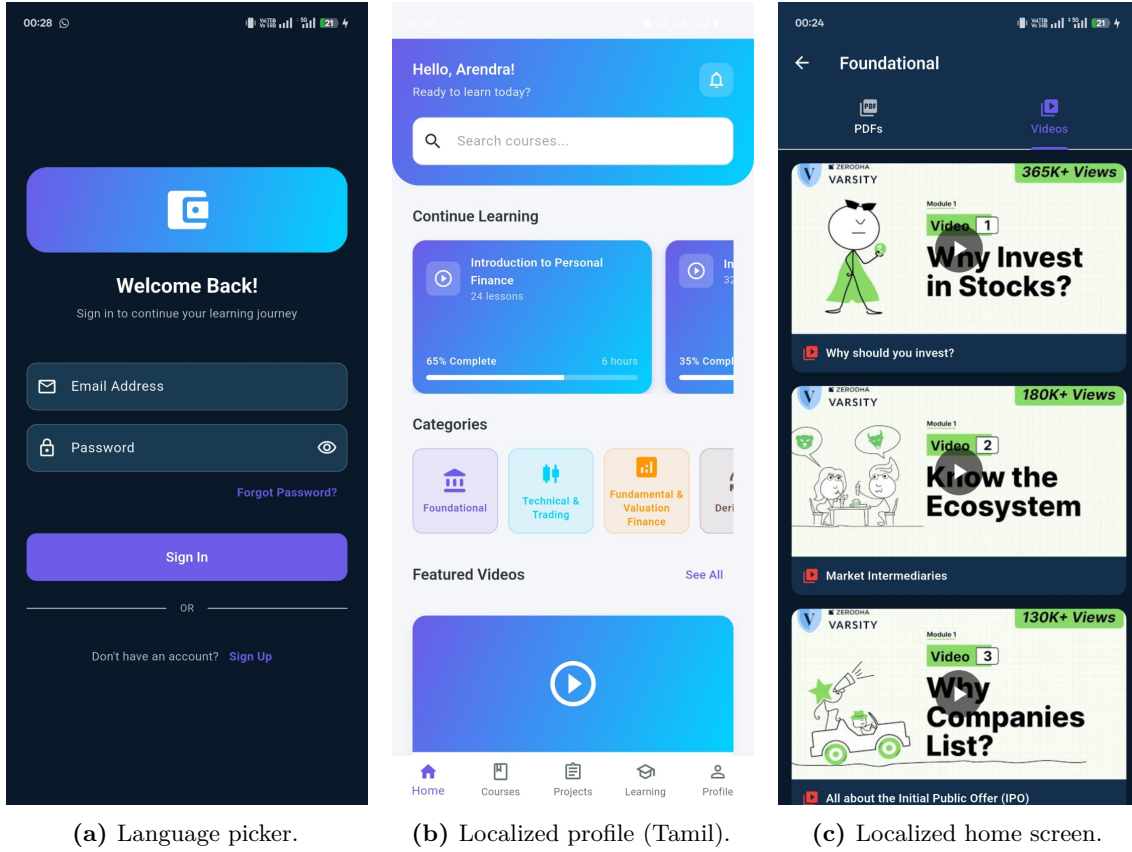


Figure 12. Finlearn interface snapshots showing onboarding, personalized dashboard, and foundational course navigation.

6.3 Structured Learning Paths

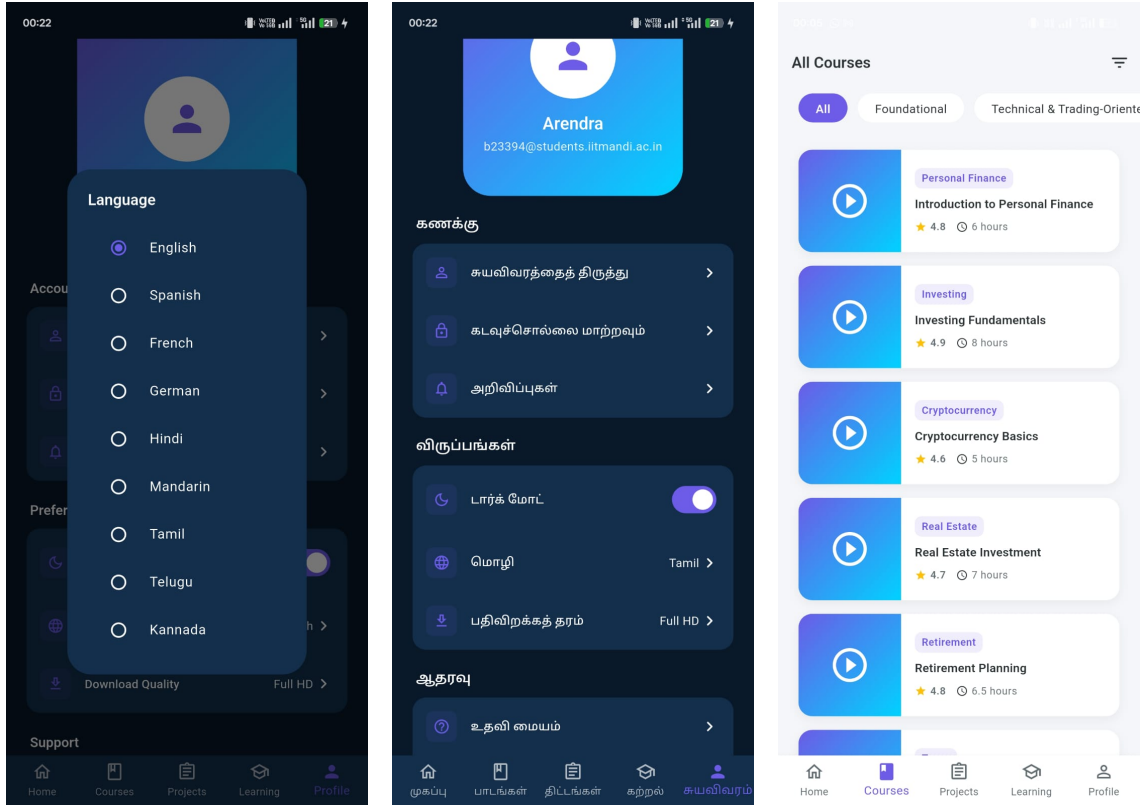
Users can track their learning progress and browse through category-wise topics, allowing them to navigate the content at their own pace and area of interest.

6.4 Content-Rich Learning

The platform combines high-quality video lessons with downloadable PDF resources and detailed background and lesson insights, giving learners multiple formats to absorb financial concepts.

6.5 Multi-Lingual Support

To reach India's diverse population, *Finlearn* supports more than seven languages, including Hindi, Tamil, Telugu, Kannada, Spanish, French, German, and Mandarin. The navigation surface and profile settings are fully localized, making the app accessible for non-English speakers across different cultures.



(a) Language picker.

(b) Localized profile (Tamil).

(c) Localized home screen.

Figure 13. Multi-lingual support: users can navigate the entire app in their preferred language, with fully localized navigation and profile settings.

6.6 Future Roadmap

We are continuously evolving the platform. The next phase includes:

- **Agentic Chatbot:** A dedicated chatbot agentic helper to answer specific financial and market-related questions.
- **Dynamic Chapter Updation:** A backend system to push new chapters and updated content instantly to users.
- **Dummy Portfolio Management:** A simulated portfolio management system for users to practice managing investments without any financial risk.
- **Interactive Games and Quizzes:** Engaging games and quizzes to reinforce financial concepts in a fun and effective way.

7 Project Outcomes and Recommendations

7.1 Concrete deliverables

1. **Reproducible monthly Indian macro panel** (2004 2025) with publication lag handling and a transparent imputation policy.
2. **Eight trained models** (Probit, Logit, L1-Logit, Random Forest, Gradient Boosting, AR(1) growth benchmark, base-rate prior, BiLSTM AE Attention) with frozen training-only thresh-

olds and isotonic recalibrated out of fold probabilities.

3. **End-to-end Python pipeline** ($\approx$ **3,000** lines) that runs from raw spreadsheet to publication-grade figures in a single command.
4. **15+ high-DPI figures** including ROC, PR, calibration, threshold-tuning, lead-time, feature-importance, executive dashboard, and the IRRI composite.

8 Conclusion

The exercise demonstrates three things. First, a calibrated monthly recession probability for India can be produced today using openly-available data, with best validation discrimination at PR-AUC **0.575** (Probit) and PR-AUC **0.563** (BiLSTM AE Attention), best held-out calibration at Brier **0.149** (Random Forest), and confidence intervals throughout. Second, the choice of model matters less than is often claimed what unambiguously matters is calibration, threshold tuning on training data only, lead-time evidence, and confidence intervals when the test sample is small. Third, this project is exactly what the ISTP course asks for: we built a real working tool, tested it against what people actually told us on the ground, and made everything reusable so that future teams can build on it rather than starting from scratch. Future work should focus on testing the model on live data as it actually gets released, checking how sensitive the results are to different ways of defining a recession, and talking to people across more regions of India to make sure the warning signals we picked hold up beyond Mandi.

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References

- Aastveit, K. A., Anundsen, A. K., & Herstad, E. I. (2019). Residential investment and recession predictability. *International Journal of Forecasting*, 35(4), 1790–1799. <https://doi.org/xxxxx>
- Berge, T. J., & Jordà, O. (2011). Evaluating the classification of economic activity into recessions and expansions. *American Economic Journal: Macroeconomics*, 3(2), 246–277. <https://doi.org/xxxxx>
- Chauvet, M., & Hamilton, J. D. (2006). Dating business cycle turning points. In C. Milas, P. Rothman, & D. van Dijk (Eds.), *Nonlinear time series analysis of business cycles* (pp. 1–54). Elsevier.

- Chauvet, M., & Piger, J. (2008). A comparison of the real-time performance of business cycle dating methods. *Journal of Business & Economic Statistics*, 26(1), 42–49. <https://doi.org/xxxxx>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). <https://doi.org/xxxxx>
- Davig, T., & Hall, A. S. (2019). Recession forecasting using Bayesian classification. *International Journal of Forecasting*, 35(3), 848–867. <https://doi.org/xxxxx>
- Estrella, A., & Mishkin, F. S. (1998). Predicting U.S. recessions: Financial variables as leading indicators. *Review of Economics and Statistics*, 80(1), 45–61. <https://doi.org/xxxxx>
- Federal Reserve Bank of St. Louis. (n.d.). *FRED economic data series and recession indicators*. <https://fred.stlouisfed.org>
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384. <https://doi.org/xxxxx>
- Investing.com. (n.d.). *India bond yield and market historical data (10-year proxy series)*. <https://www.investing.com>
- James, A., Abu-Mostafa, Y. S., & Qiao, X. (2019). Nowcasting recessions using the SVM machine learning algorithm. *arXiv*. <https://arxiv.org/abs/1903.03202>
- National Bureau of Economic Research. (2008). *The NBER’s business cycle dating procedure: Frequently asked questions*. NBER.
- National Stock Exchange of India. (n.d.). *Nifty 50 index historical market data*. <https://www.nseindia.com>
- Nyberg, H. (2010). Dynamic probit models and financial variables in recession forecasting. *Journal of Forecasting*, 29(1–2), 215–230. <https://doi.org/xxxxx>
- Organisation for Economic Co-operation and Development. (n.d.). *OECD data indicators*. <https://data.oecd.org>
- Reserve Bank of India. (n.d.). *RBI database and policy-rate publications*. <https://www.rbi.org.in>
- Schuster, M., & Paliwal, K. K. (1997). Bidirectional recurrent neural networks. *IEEE Transactions on Signal Processing*, 45(11), 2673–2681. <https://doi.org/xxxxx>
- Vrontos, S. D., Galakis, J., & Vrontos, I. D. (2021). Modeling and predicting U.S. recessions using machine learning techniques. *International Journal of Forecasting*, 37(2), 647–671. <https://doi.org/xxxxx>
- Wang, Z., Li, K., Xia, S. Q., & Liu, H. (2021). Economic recession prediction using deep neural network. *arXiv*. <https://arxiv.org/abs/2107.10980>
- Wright, J. H. (2006). *The yield curve and predicting recessions*. Federal Reserve Board.
- Yen, M.-F., & Huang, Y.-P. (2019). A new perspective of performance comparison among machine learning algorithms for financial distress prediction. *Applied Soft Computing*, 83, 105663. <https://doi.org/xxxxx>